

Jeevana Priya Inala

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📄 [jinala.github.io](https://github.com/jinala)

Education

2015–2016 **Master of Engineering**, *Computer Science*, Massachusetts Institute of Technology, GPA – 5/5.

Advisor: Prof. Armando Solar-Lezama

Thesis: Synthesis of Domain Specific CNF Encoders for Bit-Vector Solvers

2012–2016 **Bachelor of Science**, *Electrical Engineering and Computer Science*, Massachusetts Institute of Technology, GPA – 4.9/5.

Relevant Coursework Foundations of Program Analysis; Cognitive Robotics; Machine Learning; Computer Systems Security; Database Systems; Multicore Programming; Automata, Computability and Complexity; Performance Engineering;

Publications

Nathaniel Herman, Jeevana Priya Inala, Yihe Huang, Lily Tsai, Eddie Kohler, Barbara Liskov, Liuba Shrira. **Type-Aware Transactions for Faster Concurrent Code**. *EuroSys 2016*.

Jeevana Priya Inala, Rohit Singh, Armando Solar-Lezama. **Synthesis of Domain Specific CNF Encoders for Bit-Vector Solvers**. *Under submission to SAT 2016*.

Jeevana Priya Inala, Xiaokang Qiu, Ben Lerner, Armando Solar-Lezama. **Synthesis of Recursive ADT Transformations from Reusable Sketches**. *Under submission*. Draft at <http://arxiv.org/abs/1507.05527>

Experience

Research Projects

July 2015 – Present **Synthesis of optimal constraints for SAT solver**, *M.Eng Thesis*, CSAIL, MIT.

Supervisor: Prof. Armando Solar-Lezama

In this project, we are using SKETCH (a program synthesis tool) to synthesize optimal CNF clauses from high level DAGs represented as functions. We formalized efficient specification for synthesizing these kinds of problems and used it to synthesize constraints for several boolean functions. We are now working on using this formulation for more complex functions involving bit vectors and integrating these optimal encodings directly into SKETCH.

Oct 2014 – Present **Type-Aware Transactions for Faster Concurrent Code**, CSAIL, MIT.

Supervisors: Prof. Barbara Liskov, Prof. Eddie Kohler

Implemented a software transactional object (STO) system that tracks semantic operations on data structures rather than low-level memory reads and writes. Evaluated the approach on STAMP benchmarks to illustrate that STO performs better than regular untyped transaction systems such as TL2.

Sep 2013 – July 2015 **Synthesis of Recursive ADT Transformations from Reusable Sketches**, CSAIL, MIT.

Supervisor: Prof. Armando Solar-Lezama

Implemented SYNTREC (an extension to SKETCH) that can synthesize interesting functions involving algebraic data types and pattern matching. Uses a combination of type inference and counterexample-guided inductive synthesis (CEGIS) in order to support very high-level notations for describing the space of possible implementations that the synthesizer should consider.

Internship

Summer 2014 **Software Engineering Intern**, GOOGLE INC., Mountain View, CA.

Designed and implemented a Mobile app for Content ID feature. Improved the app with mobile specific characteristics to show that the mobile app is better than the web app.

Awards

- 2015 First place in PLDI student research competition
- 2014-2015 Actifio Undergraduate Research and Innovation Scholar
- 2012 Gold medal at 13th Asian Physics Olympiad, India
- 2011 Gold medal and Best in Theory in 5th International Olympiad in Astronomy and Astrophysics, Poland
- 2012 Silver medal and Asian Girl topper in 43rd International Physics Olympiad, Estonia
- 2012 Secured rank 21 in the prestigious Indian Institute of Technology (IIT) Joint Entrance Examination

Selected Course Projects

- Nov 2015 **Learning with Wasserstein Loss in Caffe**, *Course: 6.867 Machine Learning*, MIT.
Implemented a regularized Wasserstein loss layer in Caffe. Used it to perform supervised learning on MNIST datasets and image segmentation tasks.
- Dec 2014 **Concolic Execution for Django**, *Course: 6.858 Computer Systems Security*, MIT.
Implemented a concolic execution framework for testing Django applications. Uses Z3 solver to solve for symbolic path conditions to find the failing test case.
- Jan 2013 **BatteCode**, MIT.
The 6.370 Battlecode programming competition is a unique challenge that combines battle strategy, software engineering and artificial intelligence. Implemented Bug algorithm for navigation of soldiers and other strategies such as division of soldiers into groups, group navigation and opponent strategy based strategy.